

A. LLPS

time limit per test: 2 seconds
 memory limit per test: 256 megabytes
 input: standard input
 output: standard output

This problem's actual name, "[Lexicographically Largest Palindromic Subsequence](#)" is too long to fit into the page headline.

You are given string s consisting of lowercase English letters only. Find its lexicographically largest palindromic subsequence.

We'll call a non-empty string $s[p_1p_2...p_k] = s_{p_1}s_{p_2}...s_{p_k}$ ($1 \leq p_1 < p_2 < ... < p_k \leq |s|$) a subsequence of string $s = s_1s_2...s_{|s|}$, where $|s|$ is the length of string s . For example, strings "abcb", "b" and "abacaba" are subsequences of string "abacaba".

String $x = x_1x_2...x_{|x|}$ is lexicographically larger than string $y = y_1y_2...y_{|y|}$ if either $|x| > |y|$ and $x_1 = y_1, x_2 = y_2, ..., x_{|y|} = y_{|y|}$, or there exists such number r ($r < |x|, r < |y|$) that $x_1 = y_1, x_2 = y_2, ..., x_r = y_r$ and $x_{r+1} > y_{r+1}$. Characters in the strings are compared according to their ASCII codes. For example, string "ranger" is lexicographically larger than string "racecar" and string "poster" is lexicographically larger than string "post".

String $s = s_1s_2...s_{|s|}$ is a palindrome if it matches string $rev(s) = s_{|s|}s_{|s|-1}...s_1$. In other words, a string is a palindrome if it reads the same way from left to right and from right to left. For example, palindromic strings are "racecar", "refer" and "z".

Input

The only input line contains a non-empty string s consisting of lowercase English letters only. Its length does not exceed 10.

Output

Print the lexicographically largest palindromic subsequence of string s .

Examples

input	Copy
radar	
output	Copy
rr	

input	Copy
bowwowwow	
output	Copy
wwwwww	

input	Copy
codeforces	
output	Copy
s	

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Round 127 (Div. 2)

Finished

Practice



→ Virtual participation

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[*800](#)

input	Copy
mississipp	
output	Copy
ssss	

No tag edit access

→ **Contest materials**

- Codeforces Round #127 (ru) ✕
- Разбор задач (ru) ✕

Note

Among all distinct subsequences of string "radar" the following ones are palindromes: "a", "d", "r", "aa", "rr", "ada", "rar", "rdr", "raar" and "radar". The lexicographically largest of them is "rr".

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